Spectral Entropy with AntroPy

This example demonstrates how to calculate **spectral entropy** from a signal using AntroPy.

Spectral entropy measures how the signal's power is distributed across frequencies.

- Lower spectral entropy → power is concentrated in a small number of frequencies.
- Higher spectral entropy → power is spread across many frequencies.

We will first compare a simple periodic signal with a noisy signal to understand the behavior of the feature.

Create a Simple 10 Hz Signal

We first create a simple periodic signal at **10 Hz**.

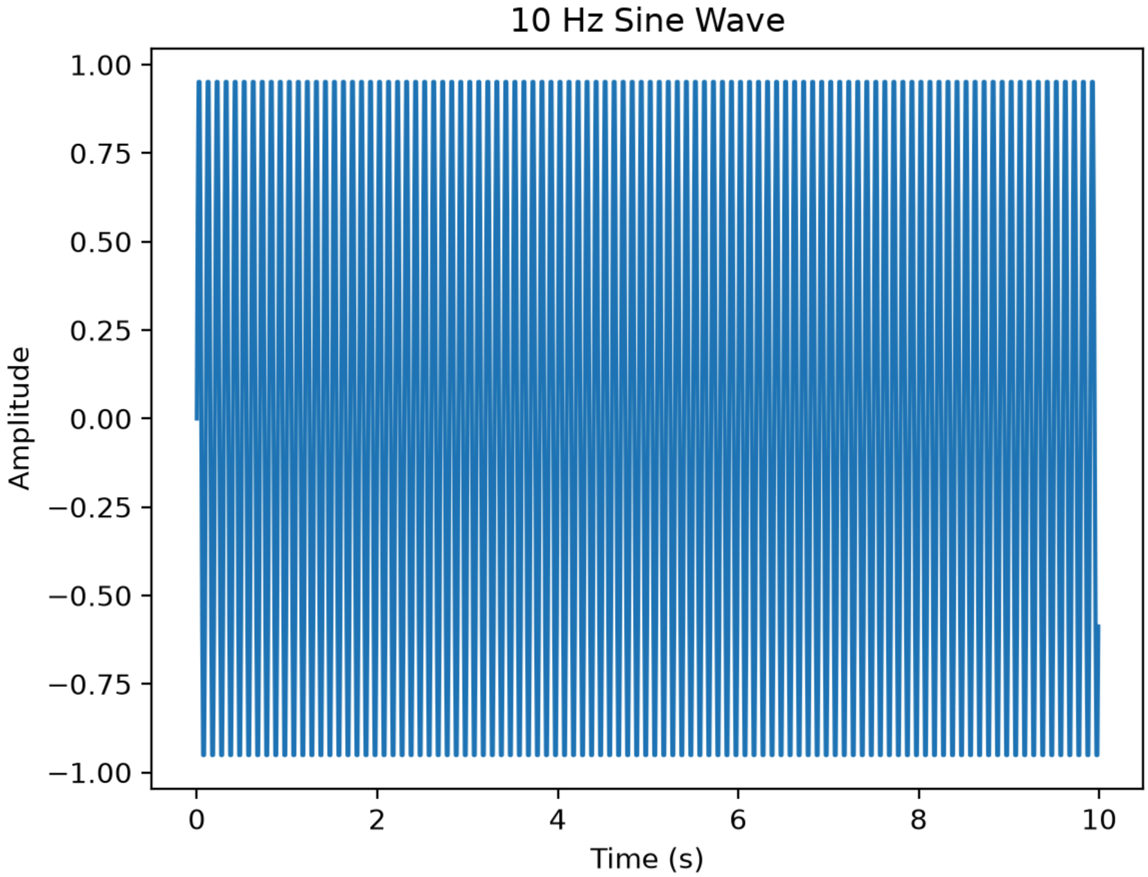
A 10 Hz rhythm is within the **alpha frequency range** commonly studied in EEG.

This signal is intentionally simple and regular, so its power is concentrated around one frequency.

Visualize the Signal

Before calculating spectral entropy, we visualize the generated signal.

Because this is a clean 10 Hz sine wave, we expect to see a regular and repeating pattern.



Create a Noisy Signal

Next, we create a random noisy signal.

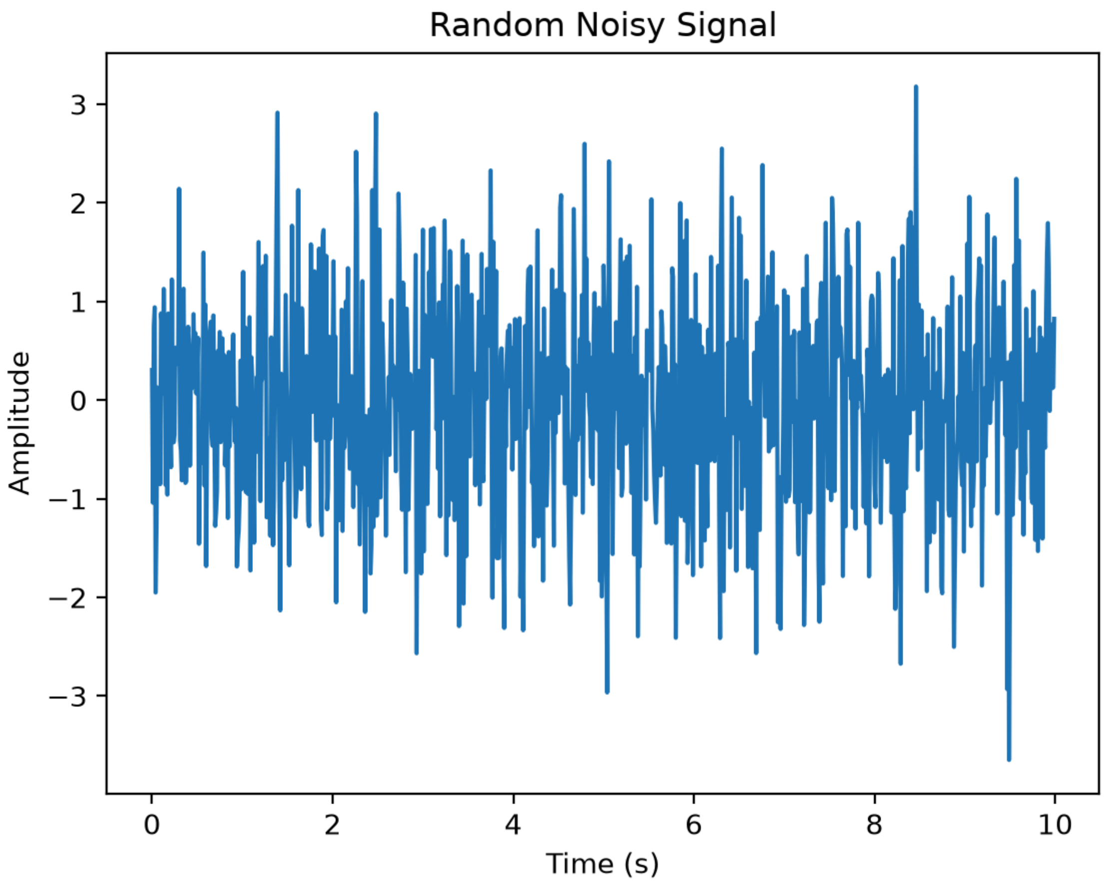
Unlike the 10 Hz sine wave, the noisy signal contains a broader range of frequency components.

We will later compare the spectral entropy of the regular signal with the spectral entropy of the noisy signal.

Visualize the Noisy Signal

We now visualize the random noisy signal.

Unlike the regular 10 Hz sine wave, this signal changes irregularly over time and does not show a clear repeating pattern.



Calculate Spectral Entropy

We now calculate the normalized spectral entropy of both signals using AntroPy.

The 10 Hz sine wave is expected to have lower spectral entropy because most of its power is concentrated around one frequency.

The noisy signal is expected to have higher spectral entropy because its power is spread across a wider range of frequencies.

Compare the Results

We compare the normalized spectral entropy of the two signals.

We expect:

- the clean 10 Hz sine wave to have lower spectral entropy;
- the noisy signal to have higher spectral entropy.

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Clean 10 Hz signal: 0.178
Noisy signal: 0.983
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